# HEPATITIS B VACCINE

Birth dose within 24 h

Healthcare workers

**CHEMICAL NAME** Recombinant hepatitis B surface antigen (HBsAg) produced in yeast, adsorbed on aluminium hydroxide/phosphate

**CLASS** Recombinant subunit vaccine (active immunisation)

**TRADE NAMES** Engerix-B · Recombivax HB · HBVaxPro · Heplisav-B

## THERAPEUTIC USES / INDICATIONS

- Universal infant immunisation (birth dose + series).
- Newborns of HBsAg-positive mothers (with HBIG within 12–24 h).
- High-risk adults: healthcare workers, dialysis, haemophilia, sexual/household contacts, people who inject drugs, travellers.
- Post-exposure prophylaxis (needlestick) in non-immune.

## ADMINISTRATION

- **IM** deltoid (adults) or anterolateral thigh (infants) — not buttock (poor response).
- Store at 2–8 °C; do not freeze; shake well.

## DOSAGES

- **Infants:** birth dose (within 24 h), then at 6, 10, 14 weeks (or per national schedule — often in hexavalent vaccine).
- **Adults:** 20 mcg (1 mL) at 0, 1 and 6 months (accelerated 0, 1, 2, 12 months).
- **Children/adolescents:** 10 mcg (0.5 mL).
- **Dialysis/immunocompromised:** 40 mcg dose.
- Check anti-HBs 1–2 months after course in healthcare workers (≥10 mIU/mL = protected).

## PHARMACOKINETICS

- Protective antibodies (anti-HBs ≥10 mIU/mL) in >95% of healthy infants/young adults after full course; lower with age, obesity, smoking, immunosuppression.

## NURSING MANAGEMENT

- Give birth dose promptly; for HBsAg-positive mothers, also give HBIG at a different site.
- Record batch, site, date; maintain cold chain.
- Healthcare workers: check post-vaccination titre; non-responders → repeat course/booster.
- Observe 15 min after vaccination.

## PHARMACODYNAMICS

- HBsAg stimulates **anti-HBs antibody** production → neutralises hepatitis B virus → prevents infection, chronic hepatitis, cirrhosis and hepatocellular carcinoma.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Injection-site pain, redness.
- Fatigue, low-grade fever.
- Headache.

### ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis (rare); yeast allergy.
- No proven link with multiple sclerosis or autoimmune disease.

## CONTRAINDICATIONS & CAUTIONS

- Anaphylaxis to previous dose or yeast.
- Severe acute febrile illness (postpone).

### KEY INTERACTIONS

- **Immunosuppressive therapy:** reduced response.
- Can be co-administered with other vaccines (different sites).

## PATIENT EDUCATION

- Complete all doses for full protection.
- Mild soreness or fever is common.
- Babies of mothers with hepatitis B need vaccine and immunoglobulin at birth, then testing at 9–12 months.

## KEY POINTS

- Recombinant HBsAg vaccine.
- HBIG + vaccine for babies of HBsAg+ mothers.
- Birth dose within 24 h + infant series.
- Check anti-HBs ≥10 in healthcare workers.